STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 11: Thermodynamics

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. State the zeroth law. [1]
2. Define internal energy. [1]
3. First law with W done by system? [1]
4. Meaning of adiabatic? [1]
5. Meaning of isothermal? [1]
6. Can a cyclic heat engine be 100% efficient? [1]
7. Ideal gas equation? [1]
8. Carnot efficiency? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. At constant volume, 500 J heat added: work? [2]
10. Gas absorbs 800 J and does 300 J work: DeltaU? [2]
11. Carnot T_h=600K,T_c=300K: efficiency? [2]
12. Ideal-gas isothermal process: DeltaU? [2]
13. 1 mol with C_v=20 J/mol K and DeltaT=5 K: DeltaU? [2]
14. Isothermal versus adiabatic? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. Gas absorbs 1000 J and does 400 J work: DeltaU? [3]
16. Carnot T_h=500K,T_c=300K: ? [3]
17. Isobaric gas expands .01 to .03 m^3 at 100 kPa. Work? [3]
18. Why no real cyclic engine converts all heat to work? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Explain reversible and irreversible processes with examples. [4]
20. Engine absorbs 1500 J from 600K, rejects to 300K; actual work=500J. Find Carnot limit, actual efficiency and Qout. [4]

End of paper - check working, units, signs and significant figures.